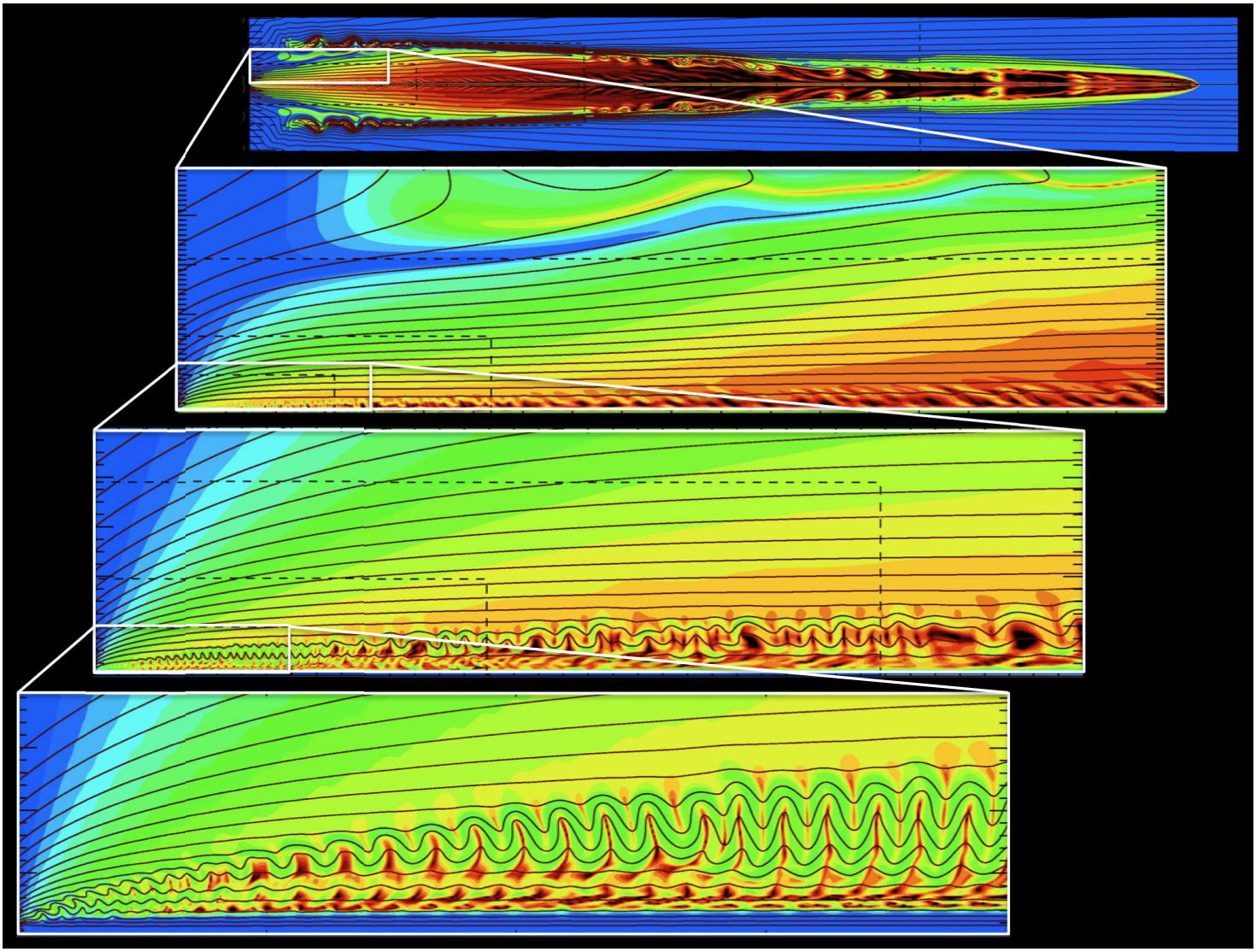
AZEuS simulation of a protostellar jet launched from a Keplerian disc



Nested grids (dashed lines) from an AZEuS simulation of a protostellar jet (flowing left to right) launched from a rotating Kelperian disc (maintained as sub-slow boundary conditions along the left boundary) showing the Alvénic Mach number $M_A = v\sqrt{\rho}/B$ (colour) superposed with black lines of magnetic induction, $\vec{B}$. As usual, ρ is the density and $\vec{v}$ is the velocity. The bottom grid represents a region $3.8\,\mathrm{AU} \times 16\,\mathrm{AU}$, while the top grid represents a region $400\,\mathrm{AU} \times 3{,}000\,\mathrm{AU}$.

This was the first successful simulation of AZEuS, and the first anywhere to track a protostellar jet from its launch site (at scales < 0.1 AU) to where astronomers routinely observe them (at scales $>$ 1,000 AU). Until this simulation, the longest reported jet simulation was 100 AU. At observational scales, we see the familiar bow-shock surrounding a magnetically dominated "nose-cone" leading the supersonic jet through the weakly magnetised ISM. At the small scales, the jet flow is unstable to the formation of knots (unrelated to HH objects) which, if observed, could give astronomers indirect local measures of temperature and magnetic field strength.

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